Welcome to SURV/SURVMETH 720 !

Total Survey Error & Data Quality 1

Class 1

August 25, 2025

Today’s Agenda

- Diverse perspectives on survey research

- Overview of TSE framework

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Diverse perspectives on survey research

Different Uses of Survey Data

- Those who describe characteristics of a finite population at a point in time ("Describer").

- Those who look for causes and correlates of phenomena in society ("Modeler").

Different Perspectives on Error

- Those who focus on sampling error and those who focus on non-sampling error.
  - Sampling statisticians : focus on errors that arise because only a subset of the population is measured.
  - Data collection and substantive experts : focus on addressing nonresponse and measurement problems.

Different Goals with Respect to Survey Error

- Efforts to measure errors
  - Measurer : design a nonresponse bias study with an intensive data collection protocol

- Efforts to reduce errors
  - Reducer : improve response rates by changing visit protocols

I find myself to be in either/both camps depending on the situation, e.g., https://www.demographic-research.org/volumes/vol50/32/50-32.pdf

Differences between Statistical and Social Science Approaches

- Statistics offers models and procedures to estimate and adjust for survey errors

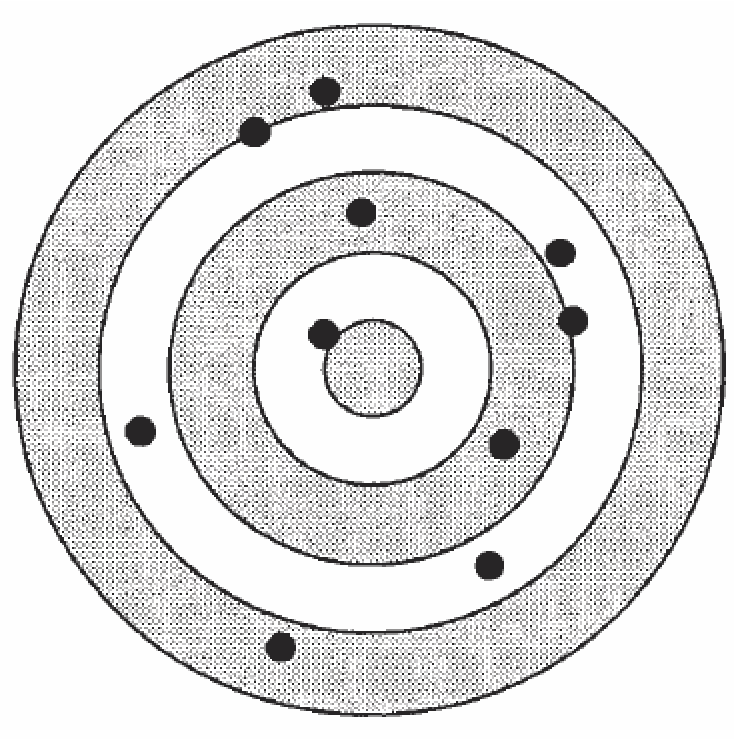
- Social science offers theories for causes of human behavior producing errors, and possible ways to address the errors

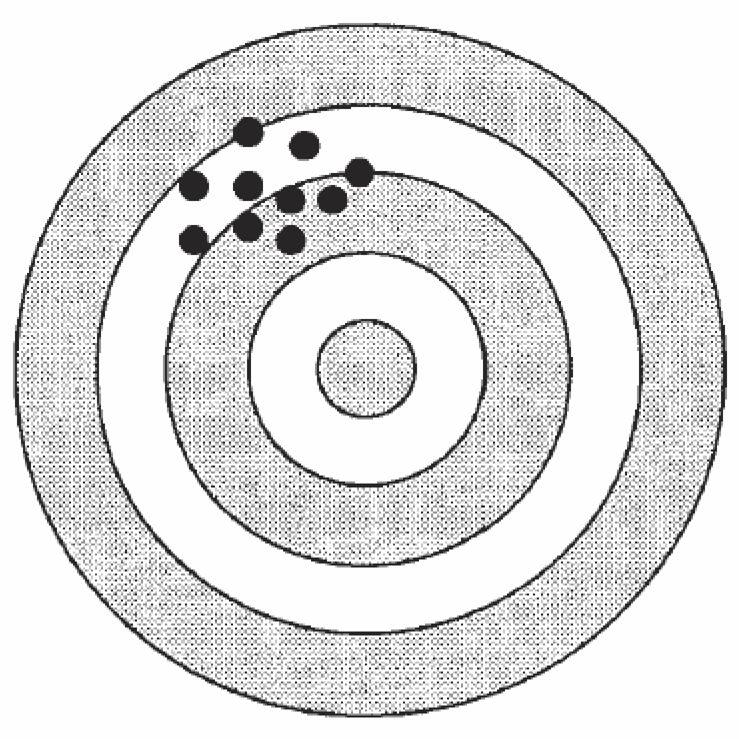
Again, I am in both ‘camps’ depending on the situation

Today's Agenda

- Diverse perspectives on survey research
- Overview of TSE framework

What do you mean by the phrase "data quality" ?





Design 1

Design 2

What would you choose ?

Building Intuition about "Survey Error"

- Let's say we are interested in a population parameter (θ).

- That parameter is unknown, and so needs to be estimated, for our purposes, using a survey ($\hat{\theta}$).

- How do we know that the survey worked well and gave us an accurate result?

- We need a metric…

Numeric Representations of Accuracy

- **Signed error**: the expected difference between an estimate ($\hat{\theta}$) and the parameter it's intended to estimate (θ)
- **Mean squared error**: the expected squared difference between θ and $\hat{\theta}$

$$MSE = \mathrm{E}\left(\hat{\theta} - \theta\right)^2$$

- Sometimes **root mean squared error** is used so the units are the same as the quantity being estimated

Survey error…

- How would you respond …?
  - "That survey was really accurate."
  - "The margin of error for my survey is _____."

- MSE is the **expected** squared difference between θ and $\hat{\theta}$.
  - Same survey process is repeated many times for the same population and under the same survey conditions.

- Compute MSE if θ = 10. Is it…

  i) < 5

  ii) 5-15

  iii) >15

  ?

Survey iteration			
1	12.2		
2	6.4		
3	5.5		
4	16.8		
5	15.1		
6	2.8		
7	6.1		
8	15.3		
9	8.2		
10	13.1		
		MSE =	?

Biemer and Lyberg (2003)

MSE…

- MSE can be expressed in terms of the squared bias and variance:

$$MSE = \mathrm{E}\left(\hat{\theta} - \theta\right)^2 = Bias^2\left(\hat{\theta}\right) + \mathrm{Var}\left(\hat{\theta}\right)$$

- Bias: errors that tend to agree; non-compensating errors; systematic errors
- Variance: 'spread', errors that tend to disagree; compensating errors; variable errors

MSE cont'd

- $$MSE = Bias^2(\hat{\theta}) + \mathrm{Var}(\hat{\theta})$$
- An accurate estimate requires both small bias and small variance
- Can MSE = 0 ?

- θ = 10. Bias = ?

- Calculation: difference between the expected value of the survey estimate and the population parameter (common to all possible replications of the survey under the same design)

Survey iteration	
1	12.2
2	6.4
3	5.5
4	16.8
5	15.1
6	2.8
7	6.1
8	15.3
9	8.2
10	13.1
	10.15
	Bias = ?

- variance = ?

- Calculation: average squared difference between each survey estimate (over conceptual replications) and the expected value of the survey estimate

Survey iteration			
1	12.2		
2	6.4		
3	5.5		
4	16.8		
5	15.1		
6	2.8		
7	6.1		
8	15.3		
9	8.2		
10	13.1		
	10.15		
		Variance =	

MSE Calculations

- $$\begin{aligned} MSE &= \text{Bias}^2 + \text{Variance} \\ &= 0.15^2 + 21.81 \\ &= 0.02 + 21.81 \\ &\sim 21.83 \end{aligned}$$

This matches our calculations from slide 14 when we focused on the average value of the squared deviations (without separately calculating variance and bias).

MSE Approximations

- MSE is seldom used, in part, because estimation is difficult.

- Also, estimation requires replication with same *essential survey conditions.*

- Combining error information for two or more error components (that one can estimate given the data at hand)
  - Typically: estimate of bias from one error source such as nonresponse plus estimate of sampling variance for one iteration of a survey

"It is not to be inferred from the foregoing material that there are grounds for discouragement or that the situation is entirely hopeless with regard to the attainment of useful accuracy. My point is that the accuracy supposedly required of a proposed survey is frequently exaggerated—is in fact often unattainable—yet the survey when completed turns out to be useful in the sense of helping to provide a rational basis for action."

- Deming, 1944

Approximations of MSE are Still Useful

- To compare the accuracy of alternative data collection/estimation methods
- To optimize the allocation of resources for the survey design (especially to reduce nonsampling error)
- To provide information to data users regarding the quality of reported estimates

Building Intuition about "TSE"

- Let's say we want to optimally allocate available resources to minimize MSE for our estimate
  - increase sample size? respondent incentives? improve IWER training?
- Need way of decomposing error into component parts so can target major sources of error
- An error framework is useful…

- TSE is a conceptual framework describing statistical error properties of sample survey statistics (Groves and Lyberg, 2010)

- TSE refers to the accumulation of **all** errors that may arise in the design, collection, processing, and analysis of a survey" (Biemer, 2010)

- The TSE paradigm provides a theoretical framework for optimizing surveys by maximizing data quality within budgetary constraints. (Biemer, 2010)

- The TSE paradigm is a valuable tool for understanding and improving survey data quality because it summarizes the ways in which a survey estimate may deviate from the corresponding parameter value. (Biemer 2016)

- TSE is the sum of all the myriad ways in which survey measurement can go wrong (Smith, 2005)

- The total survey error approach is often discussed in terms of a trade-off between survey errors and costs. However, a broader statement of this approach emphasizes the several different types of survey error, inevitable survey-related effects, and the full set of cost, time, and ethical considerations that constrain research (Weisberg, 2009)

- The TSE approach emphasizes the trade-offs that must be made in trying to minimize those possible errors within the constraint structure of the available resources (Lyberg and Weisberg, 2016)

TSE…

- Basic concept goes back as far as 1944 (Deming, ASR paper).
- TSE links survey steps to major error sources.
- TSE applicable to design, implementation, and evaluation phases of a survey.

Different Categories of Errors

- Kish (1965) : Sampling vs. Nonsampling
- Groves (1989): Observation vs. Non-observation
- Lyberg and Weisberg (2016) : correlated vs uncorrelated errors
- Groves et al. (2004): Measurement vs. Representation

TSE Diagram

Measurement

Construct

Measurement

Response

Edited Response

Representation

Target Population

Sampling Frame

Sample

Respondents

Post-survey Adjustments

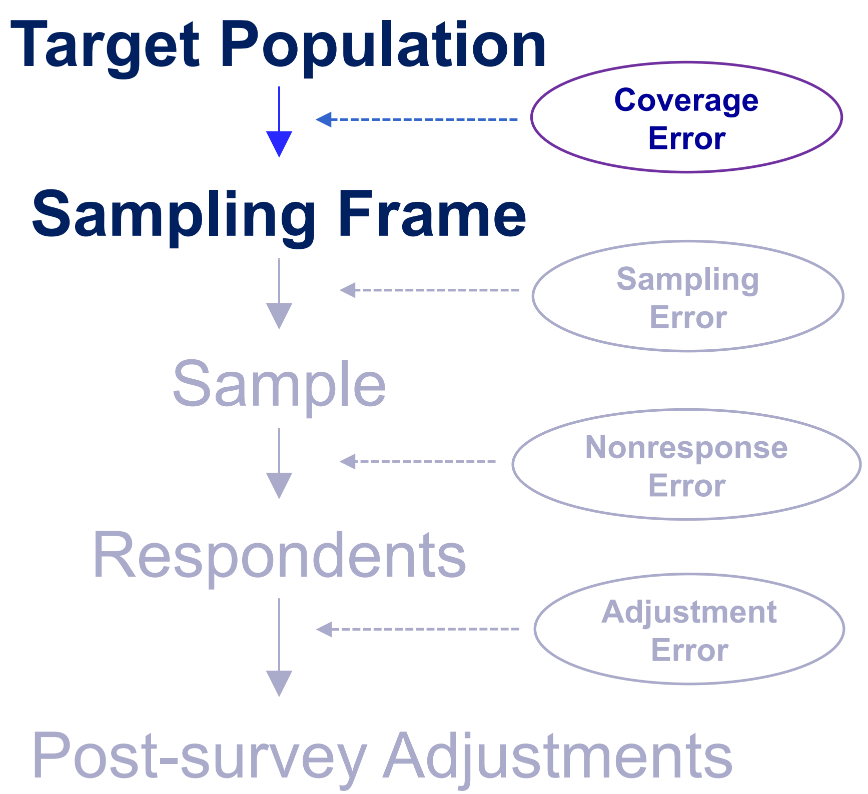
Survey Statistic

Groves et al. (2004)

Coverage Error

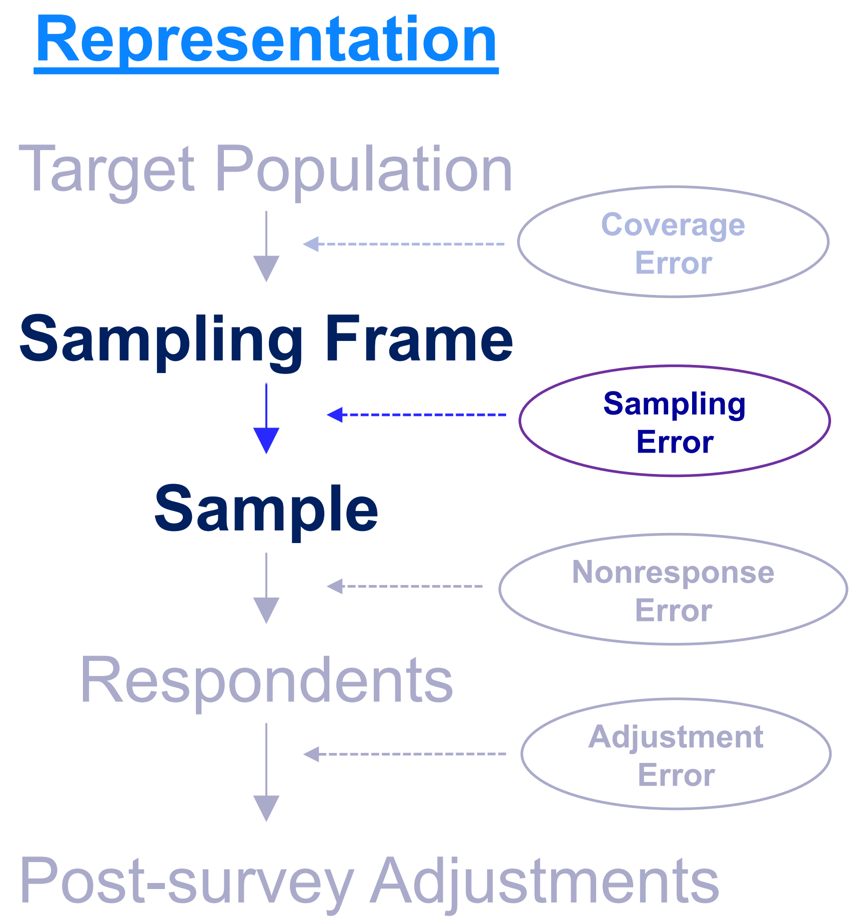
Failure to include all elements of the target population in the sampling frame

Representation

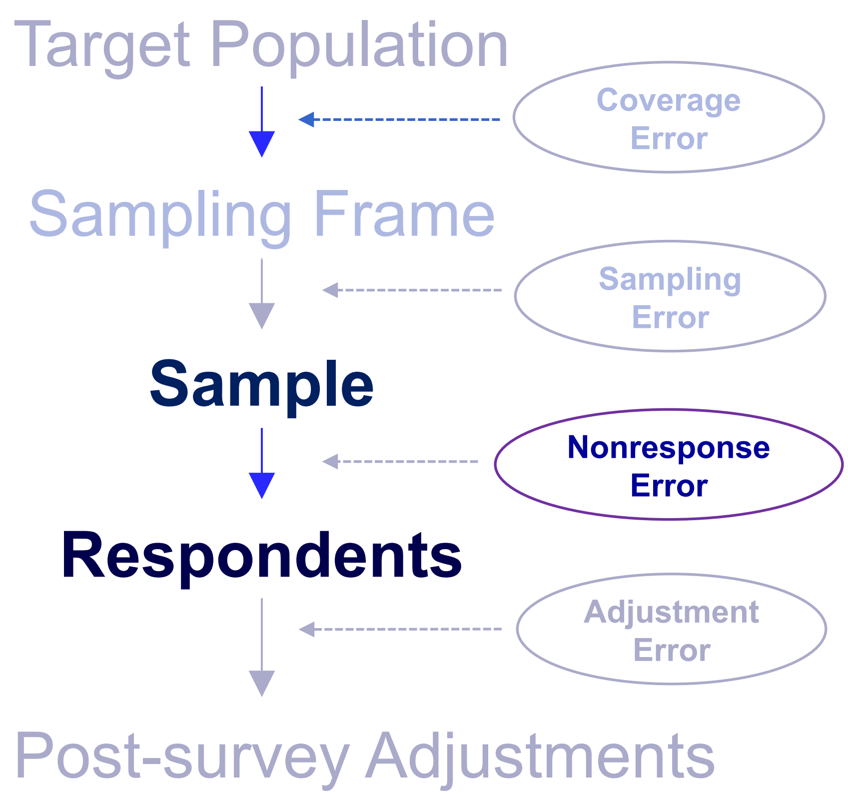


Sampling Error

Failure to include all elements of the target population

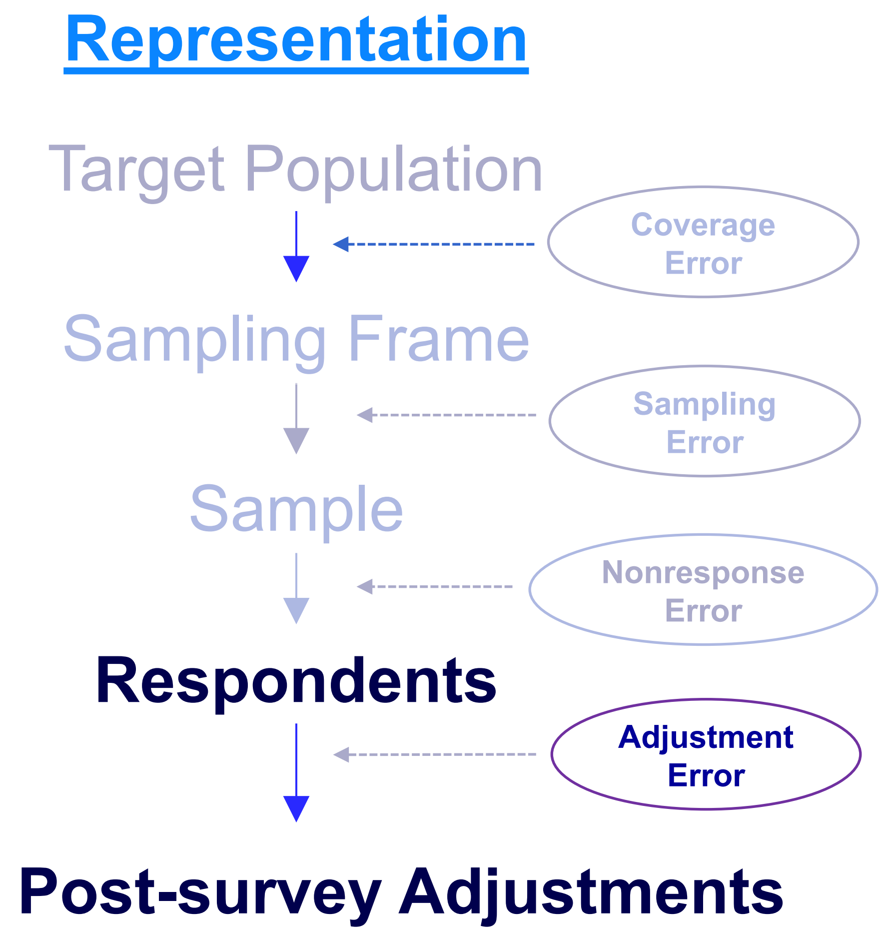


Representation



Nonresponse Error

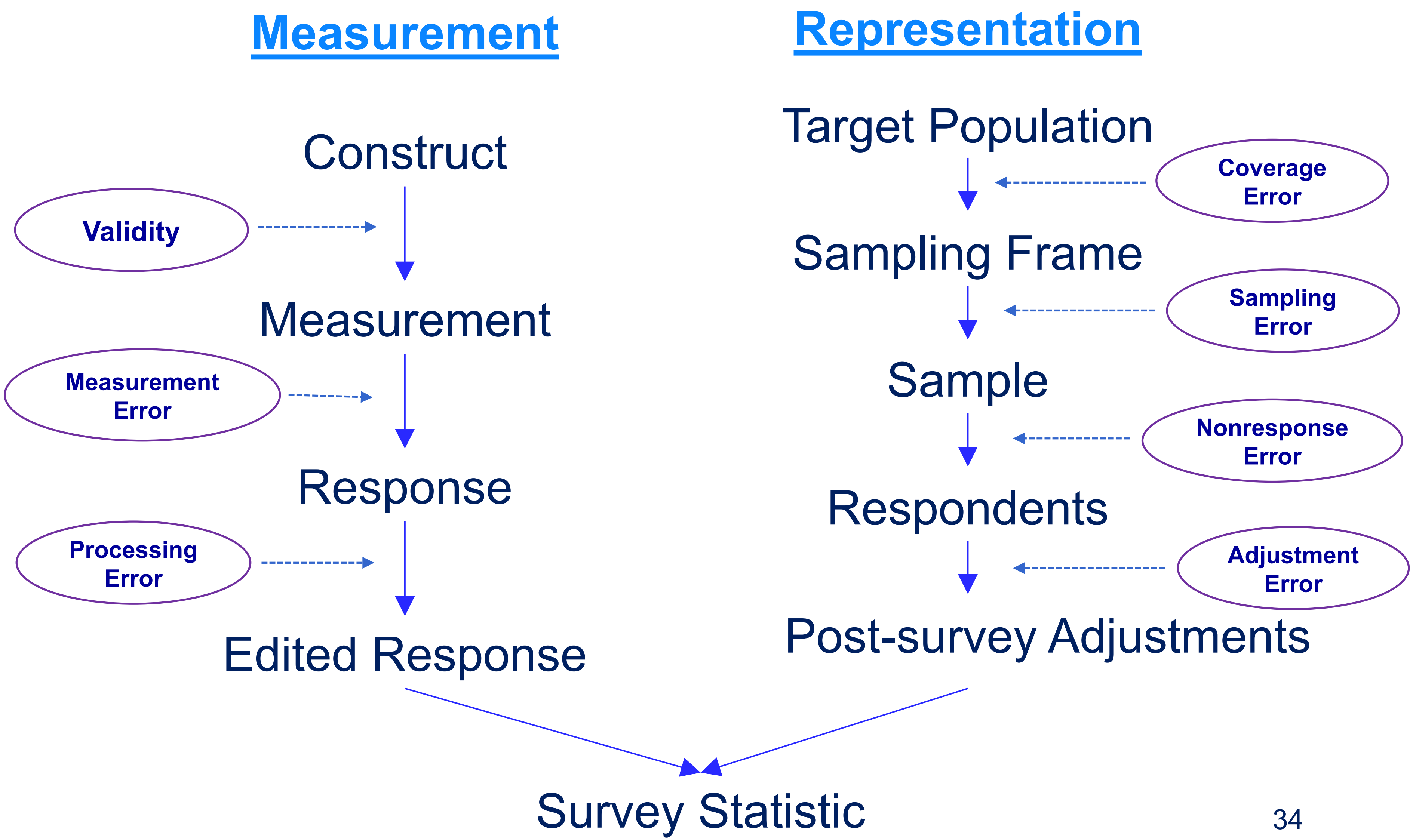
Failure to measure all selected sample members



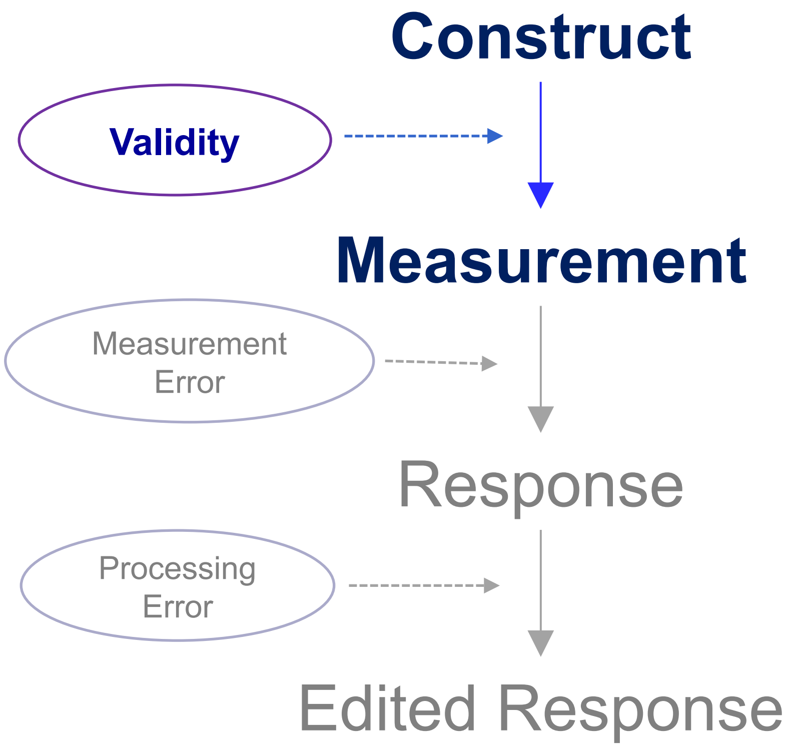
Adjustment Error

Arises from post-survey statistical adjustments (weighting, imputation, tabulation)

TSE Diagram


Measurement
Representation
Construct
Validity
Measurement
Measurement Error
Response
Processing Error
Edited Response
Target Population
Coverage Error
Sampling Frame
Sampling Error
Sample
Nonresponse Error
Respondents
Adjustment Error
Post-survey Adjustments
Survey Statistic

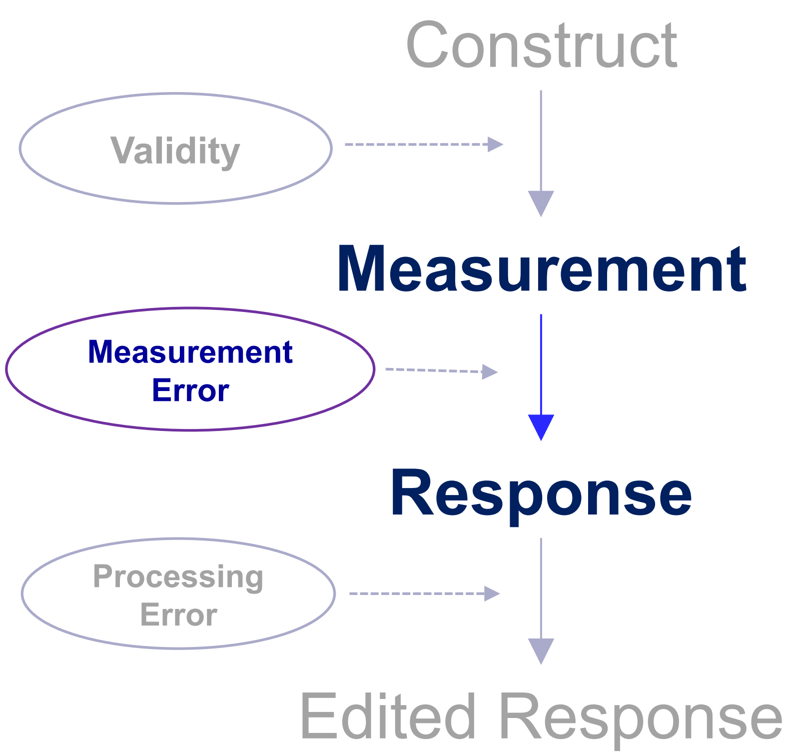
Measurement



Validity

Extent to which the measure reflects the target construct

Measurement

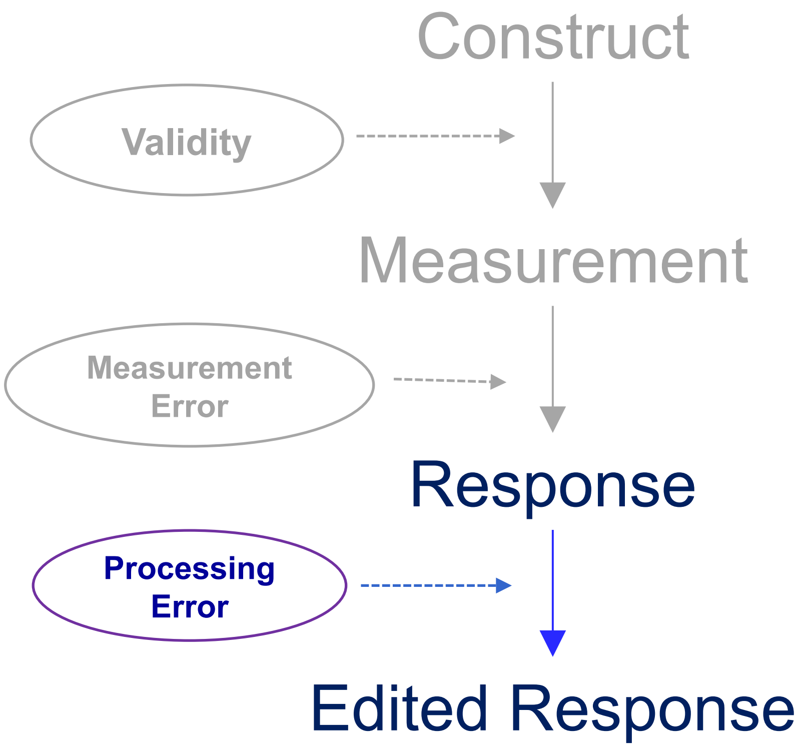


Measurement Error

Extent to which the response reflects the "true" value.

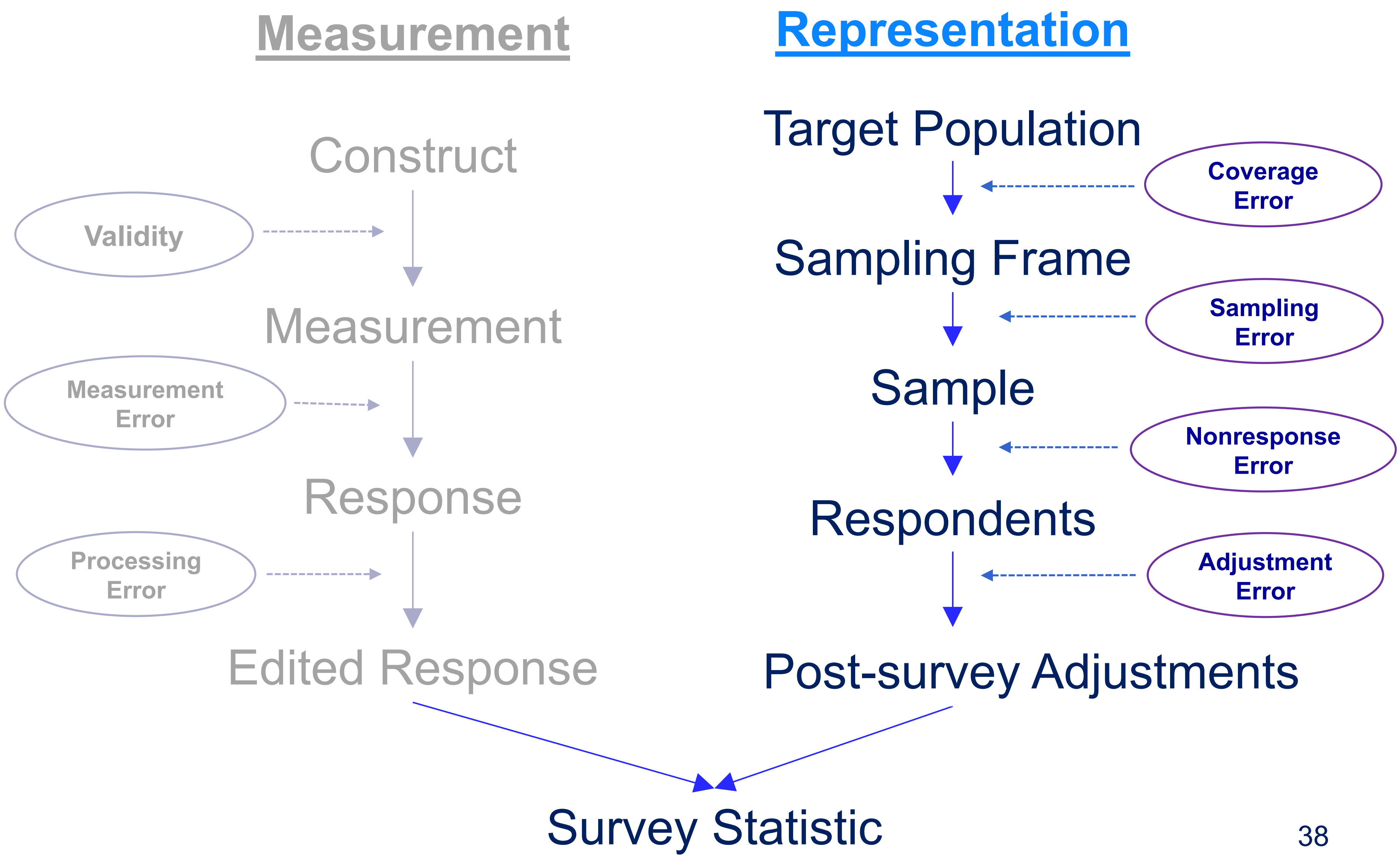
- Interviewer
- Respondent
- Instrument
- Mode

Measurement



Processing Error

Arises from processing activities that follow data collection (data entry, editing, coding)


Measurement
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Response
Processing Error
Edited Response
Target Population
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Sampling Frame
Sampling Error
Sample
Nonresponse Error
Respondents
Adjustment Error
Post-survey Adjustments
Survey Statistic

There could be more/different error sources in your application.

Group activity

Conceptually, can all sources of error that are normally associated with bias also be associated with variable error?

i) Yes

ii) No

Group Activity - Instructions

- 1. Go to: https://bit.ly/720_2025_Class1
- 2. Find the slide corresponding to your breakout room number
- 3. Introduce yourselves and write your names on the slide
- 4. You will have 10 minutes for this activity
- 5. Designate a volunteer who could speak about the slide when your group is called

Building Intuition about “Quality”

“I have designed and implemented a survey that produces very accurate estimates.”

Is this enough ?

“Fitness of Use” of Survey Statistics

Dimension	Description
Accuracy	Degree to which error is minimized
Relevance	Degree to which it meets the real needs of users
Timeliness	Delay between the reference point and the date it becomes available
Coherence	Degree to which it is consistent over time and can be successfully brought together with other information
Accessibility	Ease with which it can be obtained
Interpretability	Availability of metadata and other supplemental information necessary to interpret and utilize it

Brackstone (1999)

A System for Product Improvement, Review and Evaluation, ASPIRE

Statistics Sweden

Results for the Labour Force Survey, Round 3 in 2013

	Error Source	Average score round 2	Average score round 3	Knowledge of Risks	Communica-tion	Available Expertise	Compliance with standards & best practices	Plans or Achievement towards mitigation of risks	Risk to data quality
Accuracy(control for errors sources)	**Specification error**	70	70	◒	◒	◒	◒	◒	**L**
	Frame error	58	58	◒	◒	◒	◓	○	**L**
	Non-response error	52	52	○	○	○	○	○	**H**
	Measurement error	56	68	◒	◒	◒	○	◒	**H**
	Data processing error	62	62	○	○	◒	◒	◒	**M**
	Sampling error	78	80	◒	**O**	◒	**O**	◒	**M**
	Model/estimation error	60	64	○	○	◒	◒	◒	**M**
	Revision error	N/A	N/A	N/A	N/A	N/A	N/A	N/A	**N/A**
	Total score	60,9	64,3						

Scores					Levels of Risk			Changes from round 2	
●	◓	○	◒	**O**	**H**	**M**	**L**		
Poor	Fair	Good	Very good	Excellent	High	Medium	Low	Improvements	Deteriorations

https://www.scb.se/contentassets/edf20d31969940249b04d67cebcd5f73/ov9999_2020a01_br_x99br2101.pdf

Related concept: "Quality profile"

Recap

- TSE paradigm…
  - Part of the broader concept of Total Survey Quality (or even Total Research Quality)
  - Focuses on accuracy, that is, reducing MSE

Weaknesses of the TSE Framework

1. Exclusions of key quality concepts
   – Notably the relevance to the user

2. Diagrammatic representation might give an impression of independence of error sources

3. Lack of routine measurements
   - error/quality profiles are useful but rare
   - empirical estimation of errors is hard to achieve (but can make use of ‘effect generalizability’.); large design burden.

Weaknesses of the TSE Framework…

4. Does not define acceptable error thresholds

5. Ineffective influence on professional standards
   - press releases by many survey organizations rarely mention anything beyond sampling errors (if that)

Strengths of the TSE Framework

1. Decomposition of errors
   - nomenclature for different components

2. Separation of phenomena affecting statistics in different ways
   - measurement vs. representation issues
   - variance vs. bias
   - provides way to evaluate alternative data collection procedures (trade-offs, resource allocations)

3. Conceptual foundation of the field of survey methodology
   - subfields defined by errors

Next Class (Sep 8; enjoy the upcoming long weekend)

- Topic: Costs and coverage of target population

- Reading Assignment

- Questions?